ICES-XPS: Database Update and Modeling of Climate Impact and Adaptation Channels for the ACCREU Project

Technical Documentation

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Summary

This document describes the ICES-XPS model as used in the ACCREU project: an extended public sector version of the ICES model, with a database updated to 2017 from GTAP 11 and enriched with government and household flows, and the channels through which climate impacts and adaptation are modeled for seven hazards: sea-level rise, riverine floods, agriculture, pollination, wildfire, heat-related health and labor.

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1. Introduction

ICES eXtended Public Sector (ICES-XPS) model is an extension of the ICES core model (Eboli et al., 2010) with an explicit government institution (Delpiazzo et al., 2017). The model has not been built for ACCREU, but it has been adapted for the purposes of the project. The adaptation consisted of better characterizing the climate impact and adaptation channels in the model, and of updating the ICES-XPS database to move the base year forward and to exploit the GTAP 11 database (Aguiar et al., 2022). The model needs an enhanced database considering the economic flows between government, households and the rest of the world, which the GTAP database does not contain.

Section 2 recalls the structure of the ICES-XPS model, the extended public sector version of ICES, which splits the regional household into a private household and a government and models transfers, interest payments, aid, savings and public and private investment. Section 3 describes how the ICES-XPS database was updated to the 2017 base year, and Section 4 describes how the climate impact and adaptation channels are modeled in ICES-XPS.

2. ICES-XPS model

ICES eXtended Public Sector (ICES-XPS)(Delpiazzo et al., 2017; Parrado et al., 2020) is a global computable general equilibrium model built on the ICES core model (Eboli et al., 2010), which derives from the GTAP model (Hertel, 1997) and its energy-environment version (Burniaux and Truong, 2002). It keeps the production, trade and Armington structure of the core ICES model and extends the institutional detail of the economy with an explicit government.

The regional household of the core, which allocates regional income between private consumption, government consumption and saving, is split into a private household and a government. The private household receives factor income, transfers from the government and interest on public debt, and allocates this income between private consumption and saving. The government has its own budget, in which income from taxes, interest, international aid and other sources finances expenditure by commodity and public saving.

Private household income (YH) and government income (YG) in region r are defined as follows:

$$YH_r = \sum_e ENDOW_{e,r} + TRN_GOV_r + INTD_r + INTFI_r + \alpha\, OEXP_r - OINC_r$$

$$YG_r = TTAX_r + \sum_s AIDI_{s,r} - \sum_s AIDO_{r,s} - \sum_s INTFO_{r,s} - INTD_r - OEXP_r + OINC_r - TRN_GOV_r$$

ENDOW is the remuneration of the primary factor endowments owned by households, TRN_GOV is the government transfer to households, INTD is the interest paid on national debt to households, INTFI is the interest paid on foreign debt to domestic households, TTAX is tax revenue, AIDI and AIDO are international aid inflows and outflows, INTFO is the interest paid on debt held by foreign households, and OEXP and OINC are other government expenditure and other government income. The initial values of OEXP and OINC are tuned to match statistics on household disposable income.

In the baseline, the flows between government and households follow the evolution of income. Transfers from the government to households (TRN_GOV) and other government expenditure (OEXP) change in line with government income, and other government income (OINC) changes in line with household income. Total government expenditure moves in line with GDP. The aid paid by a region (AIDO) changes with its government income, and the aid received (AIDI) changes by the same percent in all regions, so that aid paid equals aid received in the world. In the impact and adaptation scenarios, the baseline paths of these variables are imposed as exogenous variables, to limit the channels of response to the shocks. The dynamics of interest payments (INTD, INTFI and INTFO) are described below, in the paragraphs on public debt.

Income must equal the respective final uses (total expenditure, EXPH and EXPG, and savings) for both household and government:

$$YH_r = \sum_i VDPM_{i,r} + \sum_i VIPM_{i,r} + \sum_i PSTAX_{i,r} + SAV_HHLD_r$$

$$YG_r = \sum_i VDGM_{i,r} + \sum_i VIGM_{i,r} + \sum_i GSTAX_{i,r} + SAV_GOV_r$$

VDPM and VDGM are household and government expenditure on domestic goods, VIPM and VIGM are expenditure on imported goods, PSTAX and GSTAX are the taxes on household and government consumption, and SAV_HHLD and SAV_GOV are household and government savings. The first three terms of each identity are the total expenditure of the private household, EXPH, and of the government, EXPG, that is, their purchases of domestic and imported goods and the consumption taxes they pay. Savings are the part of income that is not spent, SAV_HHLD = YH - EXPH and SAV_GOV = YG - EXPG, and country savings are the sum of household and government savings. In the impact and adaptation scenarios, government expenditure follows the exogenous path of government consumption, so that government saving is the residual of the budget. Household expenditure and saving are determined together by the allocation of household income between consumption and saving.

Net investment, that is, investment net of capital depreciation, is split into private (PINV) and public (GINV) investment, so that different accumulation rules can be specified for the two. The split is a Cobb-Douglas allocation based on the initial share of each in net investment. Public investment is financed with public saving, the difference between government income and expenditure, and with public borrowing. Net public borrowing is public investment minus public saving, PUB_BOR = INV_GOV - SAV_GOV, so a positive value means that the government borrows. Its change in each period is the change in public investment minus the change in public saving. Total saving of a region is the sum of private saving, which results from the allocation of household income between consumption and saving, and public saving.

Public debt is the stock that accumulates net public borrowing. At the end of each period, net borrowing is added to the debt, which is split into a domestic and a foreign part with a foreign share that is kept fixed over time:

$$DDGN_r = DDGO_r + (1 - FDSHR_r)\, PUB_BOR_r$$
$$FDGN_r = FDGO_r + FDSHR_r\, PUB_BOR_r$$
$$DEBT_r = DDGN_r + FDGN_r$$

where PUB_BOR is net public borrowing, DDGO and FDGO are the domestic and foreign debt of the previous period, and DDGN and FDGN are the new stocks.

Interest flows follow the stock of debt. In each period the interest due on the new stock, given the domestic interest rate DIR and the foreign interest rate FIR, is compared with the interest flow of the previous period, and the difference is imposed as a change on the flow. Three flows are updated in this way, with the same names as in the income equations: the interest paid by the government to domestic households (INTD), the interest paid abroad (INTFO), and the interest received from abroad by domestic households (INTFI). Their changes are

$$\Delta INTD_r = DIR_r\, DDGN_r - INTD_r$$
$$\Delta INTFO_r = FIR_r\, FDGN_r - INTFO_r$$
$$\Delta INTFI_r = SHRPSAVE_r \sum_s \Delta\, INTFO_s$$

where SHRPSAVE is the share of each region in the private saving of the previous period. Interest paid abroad by all regions is therefore received by regions in proportion to their past private saving. A region that accumulates public debt pays more interest to its households and abroad in the following periods, and a region that saves more receives a larger share of foreign interest.

Adaptation enters the model through two channels, both of which act on the government budget. The first is higher public expenditure: adaptation spending raises government consumption, which increases GOVEXP and reduces public saving. The second is higher public investment: adaptation capital raises the public share of net investment, which increases INV_GOV. Both channels widen the net borrowing need, so that more debt is accumulated and, since the interest flows are recomputed in each period from the updated debt stock, interest spending (INTD, INTFO, INTFI) shifts and feeds back into household and government income in the following periods.

3. ICES-XPS database construction

3.1 Data sources

The ICES-XPS database is built from the 2017 GTAP 11 database, complemented by the fiscal and household variables listed in Table 1, all expressed as a percentage of GDP and averaged over 2015–2019 to obtain the 2017 reference share. Missing values are imputed as described in Section 3.2.

Table 1. Data sources of the ICES-XPS database.

Variable	Main sources	Notes
Government saving, other government expenditure and revenue	IMF Government Finance Statistics (GFS); IMF World Economic Outlook (WEO); World Bank WDI	General, central and budgetary central government, in percent of GDP; WEO and WDI used as covariates and to fill gaps
Net lending/borrowing	IMF WEO; World Bank	2017 mean of the WEO series
Government gross debt (total, domestic, external)	IMF data tables and Country Reports; IMF WEO; World Bank International Debt Statistics; national and regional sources (ECB, ADB Basic Statistics, FRED, UN ECLAC, Statistics Canada, UNDP, ECCB)	Domestic or external debt computed by difference when only one component and the total are available; values in US dollars converted with IMF GDP
Interest payments (total, domestic, external)	IMF GFS data tables and IMF Country Reports (general or budgetary central government); World Bank International Debt Statistics (interest on external public debt); ECLAC Economic Survey of	Where only the total or the external component is available, the other is derived using the domestic/external debt shares; ECCB values are converted from XCD to US

Variable	Main sources	Notes
	Latin America and the Caribbean 2023 and CEPALSTAT; Eastern Caribbean Central Bank (ECCB); national sources for Taiwan and Liechtenstein	dollars (2.7 XCD per USD) and expressed in percent of IMF GDP
Interest receipts	Compiled from the sources above; regional means where data are missing	Missing countries filled with country-group means
Transfers	IMF GFS data tables (general, budgetary central or central government) and IMF Country Reports; World Inequality Database (WID); CEPALSTAT; P4SP database; national sources (Taiwan, India, Ethiopia, Panama, Trinidad and Tobago, Oman)	Defined as in GFSM 2014; countries with no data are filled with group means (e.g. CIS countries)
Grants received and paid by government	IMF Balance of Payments (general government credit and debit, US dollars)	Converted to percent of GDP with IMF GDP; WEO GDP per capita used as covariate
Government investment	World Bank	WEO total investment used as covariate
Household saving	World Inequality Database (WID): household saving share as first choice, and household income minus consumption as second choice; OECD household saving as third choice	IMF WEO (total investment, gross national saving, GDP per capita) and World Bank WDI (employment-to-population and unemployment rates) are used only as covariates in the imputation
Country classification	GTAP 5 to GTAP 11 regional aggregates; World Bank income group and region	Used for the hierarchical GTAP-based imputation (Section 3.3)

3.2 Imputation for missing years

The collected variables are all expressed as shares of GDP and contain missing data. To impute the missing values over time we used the MissForest method (Stekhoven and Bühlmann, 2012), a Random Forest-based technique that predicts the missing values of a variable using the information of the other variables in the dataset. Time series with fewer than two observations are excluded from the imputation. After the imputation, we compute for each variable the 2017 average share by country over the period 2015-2019.

The same procedure is applied to the government and household variables: transfers between government and households, grants in and out, debt repayments, government investment, government saving, household saving, and government gross debt with its external ownership. Each variable is imputed using its own lagged values and a set of macroeconomic covariates. The covariates come from the World Economic Outlook (WEO), all as shares of GDP: general government revenue and expenditure, gross national saving, net lending/borrowing, gross debt, total investment, the current account balance and GDP per capita. For some variables they also include the employment-to-population ratio and the unemployment rate from the World Development Indicators (WDI). The set of covariates differs by variable.

Net lending/borrowing and the inflows of debt repayments are not imputed, because their time series are complete.

Some variables are derived after the imputation as residuals. Domestic debt repayments are total repayments minus the share attributed to foreign investors, and the domestic ownership of public debt is gross debt minus external ownership. Other government expenditure is total expenses minus the components recorded explicitly (use of goods and services, interest, grants and social benefits), and other government revenue is total revenue minus tax and grant revenues. Government saving is revenues minus expenses for each level of government, and the value used is that of the general government if

available, otherwise the central government, and finally the budgetary central government.

The target household saving comes mainly from the World Inequality Database (WID), with indirect estimates when it is missing: the difference between net secondary income and private expenditure, the OECD household saving, and residual approaches based on the net saving of households and non-profit institutions serving households or of the total economy minus the other institutional sectors.

3.3 GTAP-based imputation for missing countries

The database that combines all the variables described above reports their average 2017 value as a share of GDP, after the imputation for missing years. Some countries still have missing values. For each of them, the missing share is replaced with the average share that the variable takes across the GTAP regional aggregates to which the country belongs. We then compute the monetary value of the variable for each country using the GDP of the WEO. These values, and the GDP, are aggregated by GTAP 11 region and new shares are computed. The shares are applied to the GTAP GDP values to obtain the estimates for 2017.

Some regions remain without a value after this step, for other government expenditure and revenue, government investment, and the target variables for household and government saving. The calibration of savings is unconstrained when household and government saving are missing. Government investment, other government expenditure and other government revenue are filled with the average values of the more general regional aggregates of the World Bank classification. The aim is a full database describing the new flows between agents in the economy.

3.4 First balancing of grants and interest payments

A first balancing is performed on grants and interest payments, so that at the global level inflows equal outflows. For grants, the grants in are rescaled using their share across countries and the global value of grants out. For interest payments, the interest in is rescaled in the same way using the global value of interest payments out. The change to the interest inflows is minor. The global value of grants inflows is instead around 61% of the global value of grants outflows, so balancing the grants inflows implies a moderate change with respect to the initial values.

3.5 Rebalancing of the ICES-XPS database

The balancing of the ICES-XPS database is a constrained cross-entropy minimization that adjusts the dataset to meet predefined household and government saving targets. It keeps the database internally consistent while minimizing the Kullback-Leibler divergence between the adjusted and the original shares of GDP. Household and government saving are required to fall within ±5% of their targets, and the problem is solved in GAMS with the IPOPT solver. The procedure depends on the structure of the database and on the equations chosen to represent household and government saving. Two budget constraints, expressed as shares of GDP, guide the balancing:

$$GSAVE_r = \overline{TTAX}_r + AIDI_r - AIDO_r - \overline{TRANS}_r - INTD_r \\ -INTO_r - OTH_EXP_r + OTH_REV_r - \overline{GEXP}_r \qquad (1)$$

$$HSAVE_r = \overline{FACT}_r - \overline{VDEP}_r + INTI_r + INTD_r + OTH_EXP_r$$
$$-OTH_REV_r + \overline{TRANS}_r - \overline{HEXP}_r \qquad (2)$$

Government saving in region r, GSAVE, increases with tax revenues, TTAX, international grant inflows, AIDI, and other residual revenues, OTH_REV. It decreases with grant outflows, AIDO, transfers to households, TRANS, interest payments on domestic and foreign debt, INTD and INTO, government expenditure, GEXP, and other residual expenditure, OTH_EXP. Household saving, HSAVE, increases with the remuneration of the primary factors (wages, capital returns and land rents), FACT, interest received on domestic public debt and on foreign assets, INTD and INTI, transfers from the government, TRANS, and other residual expenditure, OTH_EXP. It decreases with capital depreciation, VDEP, other residual revenues that flow to the government, OTH_REV, and household expenditure on goods and services, HEXP.

For simplicity, equations (1) and (2) and all the variables are expressed as shares of GDP in the minimization model, and all the variables refer to 2017. The variables marked with an overscore are not adjusted during the balancing: they are taken directly from the GTAP 11 database (Aguiar et al., 2022), except for TRANS, which is kept at its value in the database described in the previous sections. All the other variables belong to the database described in the previous sections.

The adjusted variables are the grant inflows (AIDI), the interest inflows (INTI), the interest on domestic debt (INTD) and the other government revenues and expenditures (OTH_REV, OTH_EXP). Transfers (TRANS) are kept fixed at their values in the database. Their starting values are those of the database described above, with grants and interest as obtained in section 3.4. Grant and interest outflows are not free: AIDO and INTO are the original shares of each region in the world outflows, applied to the new world inflows, so that world grants and interest paid always equal world grants and interest received:

$$AIDO_r = \sigma_r^{AIDO} \sum_s AIDI_s$$
$$INTO_r = \sigma_r^{INTO} \sum_s INTI_s$$
$$\sigma_r^{AIDO} = \frac{AIDO_r^0}{\sum_s AIDO_s^0}$$
$$\sigma_r^{INTO} = \frac{INTO_r^0}{\sum_s INTO_s^0} \qquad (3)$$

The new values minimize the cross-entropy, that is the Kullback-Leibler divergence, from the original shares of GDP, x^0:

$$\min_x \sum_r \sum_k x_{r,k} \ln \frac{x_{r,k}}{x_{r,k}^0}$$
$$k \in \{AIDO, AIDI, INTO, INTI, INTD, OTH_EXP, OTH_REV\} \qquad (4)$$

The minimization is subject to equations (1) to (3) and to the saving targets. In the regions where a target is available (see the data sources in section 3.1), government saving and household saving, in shares of GDP, must stay within 5% of their targets, GSAVE* and

HSAVE*. Where the WID saving is missing, the household constraint is not binding. All shares are kept positive.

4. Impact and adaptation channel modeling

The ICES-XPS model was adapted to consider ACCREU climate change impacts and adaptation measures. In the baseline, government consumption, transfers, other government flows and several efficiency trends are first obtained from a previous run and then saved as exogenous paths, so that the only differences between a climate run and the baseline are the shocks. The climate change impacts and adaption measures enter the model through five channels, summarized in sections 4.1 to 4.5, and cover 7 hazards.

4.1 Productivity of primary factors

Climate impacts are added as shifter to the productivity of each primary factor (capital, labor, land, natural resources) in each sector:

$$afe_{i,j,r} = afecom_{i} + afesec_{j} + afereg_{r} + afeall_{i,j,r} + imp_agri_{i,j,r} + imp_fire_{i,j,r} + imp_poll_{i,j,r} + imp_lab_{i,j,r} + imp_lab_flood_{i,j,r} + imp_lab_slr_{i,j,r}$$

Here afe(i,j,r) is the augmenting technical change of primary factor i in sector j of region r, in percentage change: a positive value means that less of the factor is needed to produce the same value added. The first four terms (afecom, afesec, afereg and afeall) are the baseline productivity trends, by factor, by sector, by region and by factor, sector and region together. The imp_ terms are the climate shocks, set to zero in the baseline and imposed in the impact scenarios as shifts of land productivity for crop and livestock and natural resource productivity for forestry (imp_agri), the same categories for fires on cropland and pastureland and on forest natural resources (imp_fire), pollination on land by crops (imp_poll), labor productivity under heat (imp_lab) in high and low exposed sectors, labor productivity from river floods (imp_lab_flood) and from sea-level rise (imp_lab_slr).

4.2 Stocks

Climate change impacts can alter the stocks of both non-capital and capital factors. The non-capital stocks are exogenous and are shocked directly: heat-related mortality reduces the population; heat-related shocks on labor supply reduce the employment of skilled and unskilled workers; and sea-level rise reduces the land stock. For capital, a regional shock sqk(r), related to sea-level rise, and a sector shock qoes_rfl(i,j,r), related to riverine floods, enter both the accumulation and the sectoral supply of capital:

$$VKB_r\, kb_r = 100\, NETINV_r\, time + VKB_r\, sqk_r + VKB_r \sum_j \sum_{i \in ENDWC} SHR_EC_K_{i,j,r}\; qoes_rfl_{i,j,r}$$

$$qoes_{i,j,r} = \frac{100\, NETINV_r}{VKB_r}\, time + sqk_r + qoes_rfl_{i,j,r} - endwslack_{i,r} + ETRAE_i \left[pm_{i,r} - pmes_{i,j,r} \right]$$

4.3 Efficiency in using intermediate inputs

The efficiency with which intermediate inputs are used is governed by the shifter af(i,j,r). To represent the adaptation to the heat impact on labor productivity, which changes energy use in all sectors, heating and cooling efficiency shifters are added to it:

$$af_{i,j,r} = afcom_i + afsec_j + afreg_r + afsecreg_{j,r} + afcomreg_{i,r} \\ +afall_{i,j,r} + afall_imp_heat_{i,j,r} + afall_imp_cool_{i,j,r}$$

The shifters afall_imp_heat and afall_imp_cool allow simulating the change in energy use for heating and cooling in response to climate change. In the agricultural runs, the shifter afall(Mkt_serv, i, r) is also shocked for crops, to represent the change in water use efficiency in agriculture as an adaptation to climate change.

4.4 Private and public demand

In the health runs, households shift their energy demand for cooling and heating to reduce the health impacts of heat. To account for these shifts, the single private demand shifter for market services of the ICES-XPS file is split into one for heating and one for cooling, with the same change in the import equation:

$$qpd_{i,s} = apd_h_{i,s} + apd_c_{i,s} + qp_{i,s} + ESUBD_i \left[pp_{i,s} - ppd_{i,s}\right]$$

These shifters change the energy used by households for heating and cooling, but they do not constrain total energy demand: they add up to the previous energy demand, which then adjusts given the budget constraint and the price changes.

A shift in government expenditure characterizes one of the two channels of the adaptation response to climate change impacts: adaptation measures are financed through public investment (section 4.5) and through current public disbursement. In the latter case, two shifts are imposed on top of the baseline level. Total government expenditure is exogenous and is shifted by an amount equal to the adaptation costs. The commodity demand shifter gcns(i,r), which carries the baseline demand and the shocks, is shifted for the sectors involved, defined in each scenario file: public services and construction in the flood and sea-level rise scenarios. The same public expenditure shift is also used to model the costs of morbidity in the health impact scenarios, through a shift in health expenditure.

Government demand by commodity, qg(i,r), is then given by:

$$qg_{i,r} = gcns_{i,r} \qquad i \in PUBSERV$$

The set PUBSERV contains construction, public services and health. For these commodities gcns(i,r) carries both the baseline demand and the shock.

4.5 Public investment

In the investment equations, the public investment shifter dpginv(r) is exogenous and public investment qginv(r) is the result. In the adaptation runs the two are swapped: qginv(r) becomes exogenous and dpginv(r) endogenous, so that the model takes the change in public investment as given and adjusts the shifter to deliver it. A shock to qginv(r) is then the adaptation investment itself, for example in coastal and flood protection and in agricultural infrastructure for irrigation, and private investment adjusts to the remaining total.

Public investment, qginv(r), and private investment, qpinv(r), are determined relative to the output of the capital goods sector, qcgds(r), by the shifters dpginv(r) and dppinv(r), whose average dpinvav(r) is weighted by the public and private shares of net investment, XSHRGINV(r) and XSHRPINV(r):

$$qginv_r - qcgds_r = dpginv_r - dpinvav_r$$
$$qpinv_r - qcgds_r = dppinv_r - dpinvav_r$$

$$dpinvav_r = XSHRGINV_r\, dpginv_r + XSHRPINV_r\, dppinv_r$$

4.6 Hazard-specific shocks

Each hazard uses these shifters differently. The description below is taken from the run files of the RCP7 reference runs, and Table 2 summarizes, for each hazard, the impact and adaptation variables and how they enter ICES-XPS.

Sea-level rise has three impact channels: the loss of capital stock, which includes the costs of migration; labor productivity, which includes the temporary loss of workers who migrate; and the loss of land.

Riverine floods have two impact channels. They reduce labor productivity, for skilled and unskilled labor, and they destroy capital, through a shock to the capital stock of industrial, commercial and residential sectors.

Adaptation to both hazards is public and uses the same set of shocks. It combines two shocks: additional public investment, which represents spending on protection, and current government expenditure on construction and public services, which covers maintenance and is accompanied by a coherent shock on total government expenditure.

Agricultural impacts are yield changes of the factors used in agriculture. The run files also shock the use of market services in agriculture, together with the same public investment shock. The services shock represents the change in water use efficiency in agriculture, which is the adaptation channel.

Wildfire and pollination impacts enter as productivity changes: wildfires affect land and natural resources, and pollination affects land used for crops. No adaptation actions are considered.

Heat-related health impacts have two channels. Mortality is a shock on population, and morbidity is modeled as a shift in public expenditure, both total government expenditure and expenditure in the health sector. Adaptation is private: household demand for energy commodities for heating and cooling shifts.

Heat-related labor impacts shock labor supply, the productivity of labor in the high-heat-exposure sectors, and the efficiency with which energy is used for heating and cooling in industry, which is the private adaptation channel through energy use.

Table 2. Impact and adaptation variables and their implementation in ICES-XPS.

Hazard	Impact/adaptation	Variable	ICES-XPS implementation
Sea-level rise	Impact	Expected annual damages and migration costs	Shock on capital stock
	Impact	Expected affected population	Shock on labor productivity
	Impact	Migration	Shock on labor productivity
	Impact	Land loss	Shock on land stock

Hazard	Impact/adaptation	Variable	ICES-XPS implementation
	Adaptation	Sea dikes cost, investment	Shock on government investment
	Adaptation	Sea dikes cost, maintenance	Shock on government expenditure in construction and public infrastructure (and coherent shock on total government expenditure)
Flood	Impact	Expected annual damages, sectoral	Shock on sectoral capital stock
	Impact	Expected affected population	Shock on sectoral labor productivity
	Adaptation	Protection cost, investment	Shock on government investment
	Adaptation	Protection cost, maintenance	Shock on government expenditure in construction and public infrastructure (and coherent shock on total government expenditure)
Agriculture	Impact	Crop-specific yield	Percentage change of land productivity
	Impact	Forest yield	Percentage change of forest natural resource productivity
	Adaptation	Public investment costs, expansion and investment	Shock on government investment
	Adaptation	Public investment costs, efficiency improvement	Shock on water demand by agriculture
Pollination	Impact	Yield loss due to lower pollination potential and pollinator availability	Percentage change of land productivity
Wildfire	Impact	Cropland and pastureland loss	Percentage change of land productivity loss
	Impact	Forest land loss	Percentage change of natural resource productivity loss
Health	Impact	Mortality	Shock on population growth
	Impact	Morbidity costs	Shock on government expenditure in health (and coherent shock on total government expenditure)
	Adaptation	Residential electricity/fossil energy demand for cooling and heating	Shift in residential electricity/fossil energy demand for cooling and heating
Labor	Impact	Labor supply	Shock on labor stock
	Impact	Labor productivity (sectoral)	Shock on labor productivity (high- and low-exposure sectors)
	Adaptation	Electricity/fossil energy demand for cooling and heating by agriculture, industry and services	Change in efficiency in using electricity/fossil energy in sectors

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